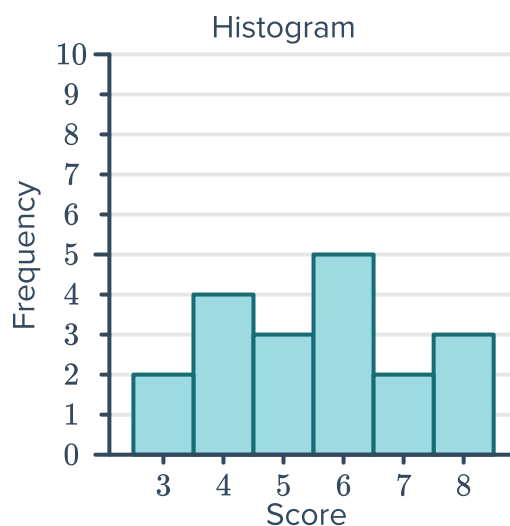


Histograms

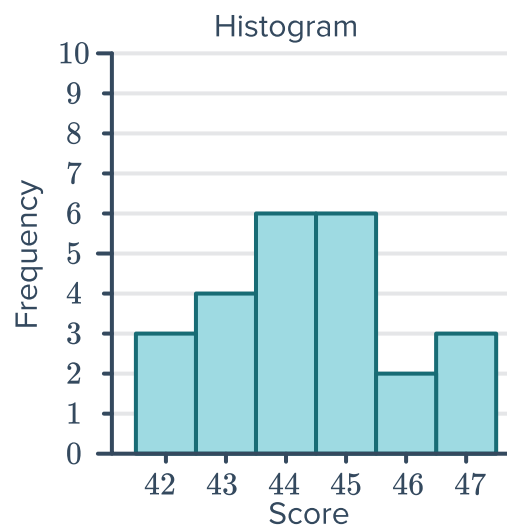


1

a



b



2

a 4

b 18

c 732

d 40.67

3

Score	Frequency
$22.5 \leq \text{Score} < 23.5$	14
$23.5 \leq \text{Score} < 24.5$	18
$24.5 \leq \text{Score} < 25.5$	16
$25.5 \leq \text{Score} < 26.5$	20
$26.5 \leq \text{Score} < 27.5$	12
$27.5 \leq \text{Score} < 28.5$	10

4

a

Time on hold in minutes	Frequency
$0 \leq \text{Time} < 1.5$	11
$1.5 \leq \text{Time} < 2.5$	12
$2.5 \leq \text{Time} < 3.5$	11
$3.5 \leq \text{Time} < 4.5$	2
$4.5 \leq \text{Time} < 5.5$	4

b 40 calls

c 17 people

d 34 people

5



a

Score	Frequency
$28 \leq \text{Age} < 30$	12
$30 \leq \text{Age} < 32$	20
$32 \leq \text{Age} < 34$	10
$34 \leq \text{Age} < 36$	11
$36 \leq \text{Age} < 38$	19
$38 \leq \text{Age} < 40$	9

b 81 mothers

c 42 mothers

d 28 mothers

6

a

Hours in Sun	Frequency
$0 \leq \text{Hours} < 10$	16
$10 \leq \text{Hours} < 20$	14
$20 \leq \text{Hours} < 30$	15
$30 \leq \text{Hours} < 40$	13
$40 \leq \text{Hours} < 50$	12
$50 \leq \text{Hours} < 60$	10
$60 \leq \text{Hours} < 70$	13
$70 \leq \text{Hours} < 80$	15
$80 \leq \text{Hours} < 90$	12

b 45 people

c $\frac{40}{120} = \frac{1}{3}$

Pie charts

7

a Pizza

b Nuggets and noodles

8

a Palpatine

b 1000 packages

9

a English

b 25%



c 128 students

10

a Russia

b China and France

c 7 countries

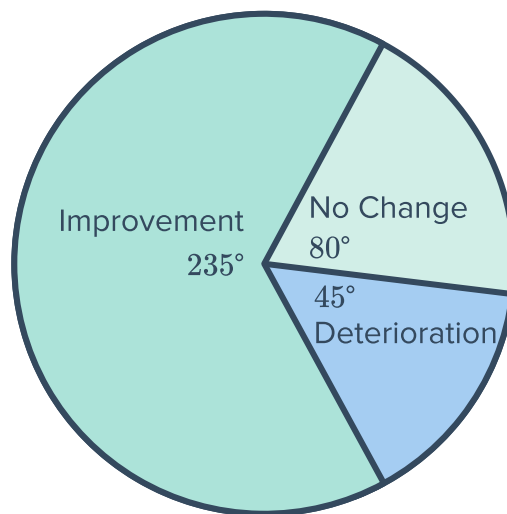
Construct pie charts

11

a

Result	Number of volunteers	Angle size
Improvement	705	$\frac{705}{1080} \times 360^\circ = 235^\circ$
No change	240	$\frac{240}{1080} \times 360^\circ = 80^\circ$
Deterioration	135	$\frac{135}{1080} \times 360^\circ = 45^\circ$

b Vaccine Test Results



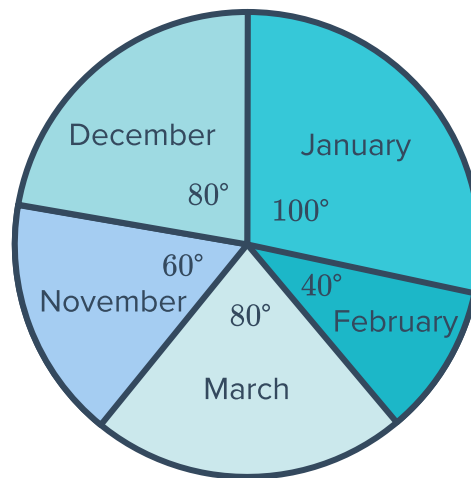
12

a

Month	Number of employees	Angle size
November	30	$\frac{30}{180} \times 360^\circ = 60^\circ$
December	40	$\frac{40}{180} \times 360^\circ = 80^\circ$
January	50	$\frac{50}{180} \times 360^\circ = 100^\circ$
February	20	$\frac{20}{180} \times 360^\circ = 40^\circ$
March	40	$\frac{40}{180} \times 360^\circ = 80^\circ$



b Holidays taken over summer



13

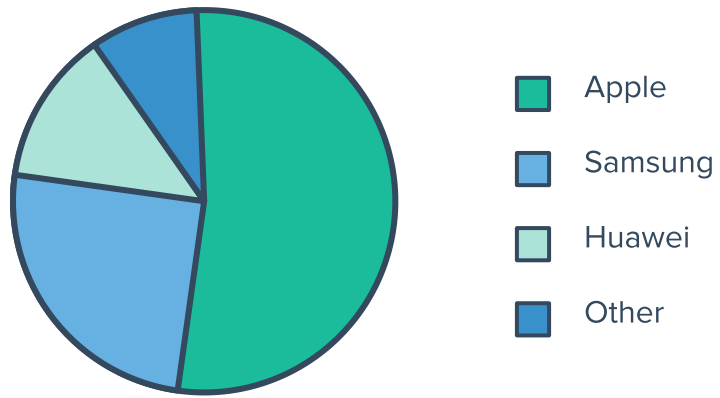
a

Manufacturer	Responses	Angle size
Apple	2051	$\frac{2051}{4000} \times 360^\circ = 184.6^\circ$
Samsung	967	$\frac{967}{4000} \times 360^\circ = 87.0^\circ$
Huawei	531	$\frac{531}{4000} \times 360^\circ = 47.8^\circ$
Other	451	$\frac{451}{4000} \times 360^\circ = 40.5^\circ$

b



Smartphone Market Share



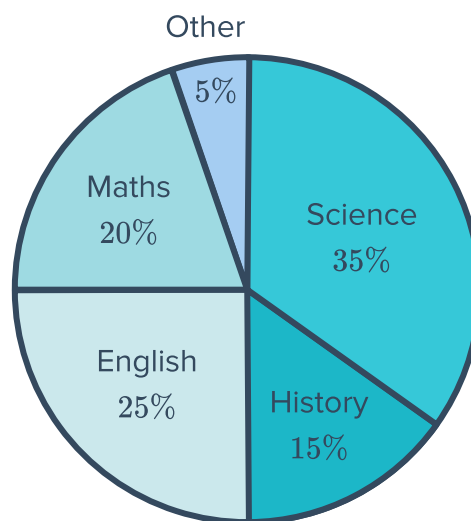
14



a

Subject	Number of students	Percentage	Angle size
Maths	44	$\frac{44}{220} = 20\%$	20% of 220 = 72°
English	55	$\frac{55}{220} = 25\%$	25% of 220 = 90°
History	33	$\frac{33}{220} = 15\%$	15% of 220 = 54°
Science	77	$\frac{77}{220} = 35\%$	35% of 220 = 126°
Other	11	$\frac{11}{220} = 5\%$	5% of 220 = 18°

b Favourite Subject



15

a The sector pieces are not the correct fraction of the circle. For example, the smallest percentage is the largest piece. Also, the sectors do not add to 100%

b B